PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Comparison of Hospital at Home Models: A Systematic Review of Reviews
AUTHORS	Leong, Man Qing; Lim, Cher Wee; Lai, Yi Feng

VERSION 1 – REVIEW

REVIEWER	Kimiko Mizuma Kagoshima University, Japan
REVIEW RETURNED	28-Aug-2020

GENERAL COMMENTS	The article entitled “Hospital at Home: A Systematic Review of Reviews” synthesize the evidence for clinical outcomes, resource use, costs, patient and caregiver outcomes and process indicator for Hospital at Home (HaH). The synthesized evidence classified according to three use cases: early supported discharge (ESD) interventions, admission avoidance (AA) interventions, and both interventions (ESD/AA) included. The authors concluded from the result that AA models should be prioritize in further explorations of the HaH. Morover, they suggested that several outcomes and process indicators still require further clarification. This review article is relevant to synthesize the evidence for the theme, since the comprehensive consolidation for the previous reviews of HaH have been required. In addition, the reviewer found several opinions and discussions about the need for HaH for COVID-19, and the reviewer think that this article is timely and important for an academic and social situation. The investigation method is appropriate, and the conclusion that the authors recommend AA model’s prioritization is unique. ESD has been better studied so far, however, this article suggested from the comparison of consolidated evidence that AA model is rather effective than ESD model. The findings from this review article will be interested from healthcare professionals in frontline and policy makers in various countries. However, the authors are required for several revisions. Major Comments ① Introduction, p5, line 10 Why the authors applied process indicators for the assessment of HaH model? Since process indicator is not familiar compared to clinical or quality indicator, the authors need to explain the definition and the need for the use in this research. ② Method, p7, line 12-14 As the implement of the assessment for the methodological quality of the primary studies is appropriate, the authors are recommended to explain the author’s original assessments criterion of risk of bias

	for enhance the method rigorous. ③ Result, p15, line 33-37 Which this LOS is, hospital LOS or total LOS? The authors are required to state which this LOS is. ④ Discussion, about LOS As mention above, the reviewer had an impression that LOS is highly important outcomes to assess HaH program's effectiveness. From the result and discussion, it is showed that total LOS in ESD and LOS (hospital or total?) in AA are increased. Is it a limitation of HaH? The authors are required to consider and discuss about this result more specifically. One of the opinions of the reviewer is that it would be interesting if the length and rate for complete cure of patients in hospital and HAH is compared. ⑤ Discussion, about ESD/AA Despite the authors showed the result of the reviews included both ESD and AA (third use case), there is no discussion about this use case. The authors also need to discuss about how you think and make implication about both included use case in Discussion. ⑥ Discussion, p19-20 4.2 Recommendation for Practice In this section, the authors are making discussion based on the previous literature rather than the result from this research. Since the discussion of HaH in the previous literature so far have been made from healthcare systems point of view, the reviewer thinks that writing the discussion based on the result of process indicator, facilitator and barrier will contribute as new opinion for HaH exploration. Minor Comments ① Introduction, p4, line 25-33 The reference should be clearly showed about the definition of ESD and AA. ② Introduction, p4, line 31-33 The explanation of AA is unclear. The reviewer cannot image clearly about AA from this explanation. ③ Method, p6, line 44-46 The several HaH schemes for specific ill was excluded from the review. The authors should show the reason for the exclusion clearly. ④ Method, p6, line 55-57 Is it appropriate to define "length of stay" including duration of home cure? Also, it is unclear whether "length of stay" is only for hospital stay or including home cure duration. The authors need to define "length of stay" more clearly. ⑤ Result, p17, line 45 The reviewer cannot understand the meaning of "proximity of the hospital". ⑦ Discussion, p18. Line 56-58 Mortality of AA is compared with readmission of ESD. The reviewer thinks comparing the result of both mortality of AA and ESD is

	appropriate in this sentence. ⑧ Discussion, p19, line 5-8 The reviewer cannot understand the meaning of the sentence from the context, “However, apart from a study in one review suggesting this, insufficient evidence regarding adverse events precluded clear conclusion”. Especially what “this” stand for? ⑨ Conclusion, p21, line 26-31 It seems that this first sentence in Conclusion is came from Introduction. The authors need to write here based on the result. ⑩ Conclusion, p21, line 31-38 The reviewer think second and third sentence recommend to write transpose, to be like the logical order written in Conclusion in Abstract.

REVIEWER	Antti Ohvanainen Siun sote, Finland
REVIEW RETURNED	30-Aug-2020

GENERAL COMMENTS	Thank you for a good manuscript on an important topic. There are yet a few things I would like you to address: 1) Hospital at home -type of services are currently being referred to by many names (hospital at home, hospital-at-home, hospital in the home, home hospital, early supported discharge services, admission avoidance and even outpatient parenteral antimicrobial therapy) How did you select the search terms you used in this study? Should the nomenclature be simpified in the future to facilitate the discovery of relevant studies? Is there already or should there be a precise MeSH for (different types of) hospitals at home? 2. You excluded i.a. reviews on palliative HAH schemes. Were these reviews about HAHs that treat exclusively palliative patients or also HAHs where palliative patients were treated among others? It would be interesting to know the number of excluded studies by group if possible. 3. The division of HAH models to ESD, AA and ESD/AA is understandable for scientific purposes, but in some case may be ill-suited for real-life settings. If an AA-type HAH is running at full capacity and is unable to admit a patient today, but can admit the same patient tomorrow (after the patient has spent one night at the hospital ward) is it actually a ESD/AA-type HAH then? A modern ESD/AA -type HAH can for example admit nursing home patients without moving the patient anywhere during the process. In general, the HAH field is in many aspects still very heterogenic and more subgroup analysis is needed in future studies. 4. You name reduction of iatrogenic events such as delirium and infections as a benefit of AA-type over ESD-type HAH. In addition to this, there is less need for repeated patient data transfer between hospital units when the patient is directly admitted to a HAH instead of after spending some time on an inpatient ward. As different ICT-systems rarely work well together and each discharge possesses a risk of relevant information being lost, the fewer possibilities of human error the better.

	5. You refer several times to the criteria of HAH, could you open this up in introduction? 6. In facilitator and barriers section you write: "In addition, caregiver availability was an eligibility criteria in one-third to half of the studies across reviews". Does this mean availability of a caregiver if one needs him/her or availability absolutely? If the latter is true, please adress the question in discussion, is the HAH designed only for people who live with someone or in a nursing home etc?

REVIEWER	Natasha Lannin Monash University, Australia
REVIEW RETURNED	03-Sep-2020

GENERAL COMMENTS	This manuscript presents the findings of a systematic review of systematic reviews of randomised trials which investigated the efficacy of hospital-at-home programs. Given the topical nature of this review during COVID-19 and a potential organisational approaches to hospital avoidance, I read this paper with great interest. Strengths of the review are that the authors rated each SR for quality using the recognised AMSTAR tool, and that the findings are discussed with respect to the three different approaches for providing HaH, and drawing on the AMSTAR ratings to provide evidence of the potential limitations of the reviews before drawing together the strength of the evidence. Details, however, are missing from the current version of the manuscript, which I believe would strengthen the paper. Main limitation is the brevity of the description of the method. It is also a limitation that the SR protocol was not registered. As I do register my own SR protocols prior to commencing, I would appreciate editorial guidance on this issue. At the least, I ask the authors to consider raising the potential biases that arise from not registering your protocol prior to commencing the review in the discussion. Further points for the authors to consider include: - AMSTAR scores could be presented as the results in the abstract, which currently lacks specificity. - details of method sufficient for replication and in accordance with acceptable guidelines are missing (for example, were reviews assessed for relevance and quality by at least two members of the review team independently?). - recommend that the authors provide an operational definition of HaH (how was the criteria of hospital-level care operationalised), and how you determined which reviews were excluded based on your criteria of "review of higher quality" - Discussion of how this paper adds to the field of evidence, considering that there are published systematic reviews with meta-analyses on this topic and your SR does not do a meta-analysed. While I appreciate that the authors report on page 7 they could not do this, the method for how the narrative summary was developed should be pre-determined and explained. The work of Prof Greenhalgh in undertaking narrative syntheses should be reflected on, and potentially the author's results section may be more readable if a theoretical model was used to synthesise their findings. The presentation of the results would benefit from greater structure so the value of synthesis across the multiple SRs is clearer to the reader.

	I would welcome reading this paper with the addition of greater detail on method, as well as presenting the results in such a way as to differentiate it out from the systematic reviews of source trials.

VERSION 1 – AUTHOR RESPONSE

This manuscript was previously submitted, reviewed and recommended for resubmission (bmjopen-2020-043285). We would like to thank the reviewers for the valuable comments. We have since gone through the suggestions in detail and our responses are as follow:

(1) *Reviewer #1: This review article is relevant to synthesize the evidence for the theme, since the comprehensive consolidation for the previous reviews of HaH have been required. In addition, the reviewer found several opinions and discussions about the need for HaH for COVID-19, and the reviewer think that this article is timely and important for an academic and social situation. The investigation method is appropriate, and the conclusion that the authors recommend AA model's prioritization is unique. ESD has been better studied so far, however, this article suggested from the comparison of consolidated evidence that AA model is rather effective than ESD model. The findings from this review article will be interested from healthcare professionals in frontline and policy makers in various countries. However, the authors are required for several revisions.*

Reviewer #2: Thank you for a good manuscript on an important topic. There are yet a few things I would like you to address.

We thank the reviewers for their kind acknowledgement. We have since gone through the comments and revised the manuscript according to the comments provided.

(2) *Reviewer #3: Strengths of the review are that the authors rated each SR for quality using the recognised AMSTAR tool, and that the findings are discussed with respect to the three different approaches for providing HaH, and drawing on the AMSTAR ratings to provide evidence of the potential limitations of the reviews before drawing together the strength of the evidence. Details, however, are missing from the current version of the manuscript, which I believe would strengthen the paper. Main limitation is the brevity of the description of the method. It is also a limitation that the SR protocol was not registered. As I do register my own SR protocols prior to commencing, I would appreciate editorial guidance on this issue. At the least, I ask the authors to consider raising the potential biases that arise from not registering your protocol prior to commencing the review in the discussion.*

We thank the reviewer for his/her valuable comments. We have since revised the manuscript according to the comments and highlighted the limitation of not registering our protocol beforehand in the discussion segment.

Abstract

(1) *Can you make it clear in the Abstract what you are referring to when you say "increased total length of stay"? Do you mean length of stay while in contact with the HaH scheme?*
Yes, total length of stay refers to the length of stay both in hospital and subsequently at home while still under the HaH scheme. The entire treatment duration under HaH schemes is considered as inpatient stay and therefore makes up the conventional "LOS". We have also clarified the definition of LOS in the abstract. (Abstract – Results)

(2) *AMSTAR scores could be presented as the results in the abstract, which currently lacks specificity.*
We have noted this suggestion and included the AMSTAR scores in the abstract. (Abstract – Results)

(3) *The Conclusion in the Abstract needs to stick with what you found; you need to tone it down to be more precise about safety, why should HaH be prioritised – on what basis, and costs for example were unclear so why should they be prioritised on this basis?*
We would like to clarify that we are recommending that AA implementations are prioritised over ESD ones in future explorations of HaH, rather than recommending the prioritisation of HaH in general. Nonetheless, we have edited the conclusion to be more precise, and to also caveat that HaH may only be suitable for certain groups of patients (Abstract - Conclusion)

Introduction

(1) *The reference should be clearly showed about the definition of ESD and AA.*
(2) *The explanation of AA is unclear. The reviewer cannot image clearly about AA from this explanation.*
We have noted these comments and clarified the definition of AA in the introduction segment (1. Introduction, paragraph 2)

(3) *You refer several times to the criteria of HAH, could you open this up in introduction?*
We have noted this comment and introduced some of the criteria of the HaH model in the introduction, such as patient eligibility based on medical condition severity and caregiver availability. (1. Introduction, paragraph 5)

(4) *Why the authors applied process indicators for the assessment of HaH model? Since process indicator is not familiar compared to clinical or quality indicator, the authors need to explain the definition and the need for the use in this research.*
Process indicators were included for data extraction because we understand that this care model is relatively new with emerging interest around the world especially since the start of the COVID-19 pandemic. Inclusion of process indicators, in our opinion, will help clinicians and hospital administrators visualise feasibility of the model in their respective settings and forecast resources required to run the service or a pilot. We have included a definition and

justification for the need for process indicators in the introduction segment. (1. Introduction, paragraph 5)

Methods

(1) *Hospital at home -type of services are currently being referred to by many names (hospital at home, hospital-at-home, hospital in the home, home hospital, early supported discharge services, admission avoidance and even outpatient parenteral antimicrobial therapy). How did you select the search terms you used in this study? Should the nomenclature be simpified in the future to facilitate the discovery of relevant studies? Is there already or should there be a precise MeSH for (different types of) hospitals at home?*

These search terms were decided or derived from existing programmes that deliver hospital-level care in the patients' homes around the world. For instance, these programmes are called "Hospital at Home" in the United States, "Hospital in the Home" in Australia, and "Home Hospital" in parts of Europe. Hence, in searching for relevant papers, we picked "Home" and "Hospital" as keywords, and looked for combinations of phrases within 3 words (i.e. NEAR/3) that had these 2 keywords. As for other search terms, they were both adapted from past systematic reviews (e.g. Cochrane), or keywords that we found important for the discovery of relevant studies. There is currently no relevant MeSH term for HaH and we concur with the reviewer that there should be a precise MeSH for this. Currently under Medline's MeSH Tree, these are the following subcategories under "Community Health Services": "Health Services, Community", "Community Health Service", "Health Service, Community", "Service, Community Health", "Services, Community Health", "Community Health Care", "Care, Community Health", "Health Care, Community", "Community Healthcare", "Community Healthcares", "Healthcare, Community", "Healthcares, Community". More specifically, under "Home Care Services", the sub-terms were all pertaining to services that are not for patients' hospital level care.

(2) *Recommend that the authors provide an operational definition of HaH (how was the criteria of hospital-level care operationalised), and how you determined which reviews were excluded based on your criteria of "review of higher quality"*

We have noted this comment and added an operational definition of HaH in the methods segment. The tool used to determine "reviews of higher quality" was AMSTAR-2; this has been added into the methods segment as well. (2.2 Study Selection, paragraphs 1 and 2)

(3) *Details of method sufficient for replication and in accordance with acceptable guidelines are missing (for example, were reviews assessed for relevance and quality by at least two members of the review team independently?)*

Yes. Search terms and criteria for inclusion/exclusion were determined a priori by all authors. Test searches were carried out and results were reviewed and refined by 2 authors before finalisation of the search strategy. Literature search and shortlist of the articles were done

consistently by 1 investigator. Shortlisted articles were reviewed by 2 authors before final inclusion for data extraction. We have included these into the methods section. (2.1. Data Sources and Searches; 2.2. Study Selection)

(4) *The several HaH schemes for specific ill was excluded from the review. The authors should show the reason for the exclusion clearly.*
HaH schemes for palliative, obstetric, paediatric and mental health HaH schemes were excluded due to differences in these groups' care needs and care objectives from general medicine patients. We aspire to find an alternative care delivery model that can be more widely applied and scaled up to benefit more patients. This would also allow for ramping up of capacity in times of crisis (e.g. COVID-19). We have added this justification in the methods segment (2.2. Study Selection, paragraph 2).

(5) *You excluded i.a. reviews on palliative HAH schemes. Were these reviews about HAHs that treat exclusively palliative patients or also HAHs where palliative patients were treated among others? It would be interesting to know the number of excluded studies by group if possible.*
These reviews were for schemes that exclusively treated palliative patients (e.g. Gomes 2013 only included primary studies where majority of patients had a severe or advanced disease). Given that the care needs, clinical trajectory and care objectives of palliative patients are vastly different from other general medicine patients, we excluded palliative care studies to allow for more meaningful comparisons.

(6) *Is it appropriate to define "length of stay" including duration of home cure? Also, it is unclear whether "length of stay" is only for hospital stay or including home cure duration. The authors need to define "length of stay" more clearly.*
Yes, total length of stay refers to the length of stay both in hospital and subsequently at home while still under the HaH scheme. The entire treatment duration under HaH schemes is considered as inpatient stay and therefore makes up the conventional "LOS". We have also clarified the definition of LOS in the methods segment. (2.3. Data Extraction and Quality Assessment, paragraph 1)

(7) *As the implement of the assessment for the methodological quality of the primary studies is appropriate, the authors are recommended to explain the author's original assessments criterion of risk of bias for enhance the method rigorous.*
We thank the reviewer for his/her suggestion. We would like to clarify that we were striving to align with the Cochrane guidelines for overview of reviews, which recommends that authors extract the original assessments presented in the systematic reviews instead of repeating or updating the risk of bias assessments that have already been conducted by systematic review authors. It was also recommended in the guidelines to present the assessments in tabular summaries, and we have opted to do so in Table 2. We have added the reference to the Cochrane guidelines in the methods segment to address similar

considerations that may be raised by other readers (2.3. Data Extraction and Quality Assessment, paragraph 2)

Results

(1) *We agree this review is important and topical. However we are missing a lot of detail about the included studies: what were the schemes that were looked at? You say there's heterogeneity but we don't get a sense about what the early supported discharge schemes were and what the admission avoidance schemes were. Length of stay isn't well defined either.*

We have elaborated on the definition of ESD and AA in the introduction as well as methods segment (1. Introduction, paragraph 2; 2.2. Study Selection, paragraph 1). We would like to clarify that the heterogeneity that we highlighted does not only pertain to differences between ESD and AA, but also between implementations within each programme type. Given the large number of primary studies and the heterogeneity between implementations, it is challenging to provide more specific information regarding the schemes. Nonetheless, we thank the editors for the comment and have elaborated on some of the heterogenous features of HaH programmes under the results segment. (3.1. Review Characteristics, paragraph 2)

(2) *Result, p15, line 33-37: Which this LOS is, hospital LOS or total LOS? The authors are required to state which this LOS is.*

As this section is reporting the results of AA implementations, LOS here refers to LOS in the home under HaH since patients are not admitted into hospital. We have added this clarification into the main text. (3.3.2. AA, paragraph 3).

(3) Result, p17, line 45: The reviewer cannot understand the meaning of "proximity of the hospital".

This refers to the physical distance of the patients' homes to the hospital. We have clarified this in the main text. (3.4 Facilitators and barriers, paragraph 1)

Discussion

(1) *Discussion of how this paper adds to the field of evidence, considering that there are published systematic reviews with meta-analyses on this topic and your SR does not do a meta-analysed. While I appreciate that the authors report on page 7 they could not do this, the method for how the narrative summary was developed should be pre-determined and explained. The work of Prof Greenhalgh in undertaking narrative syntheses should be reflected on, and potentially the author's results section may be more readable if a theoretical model was used to synthesise their findings. The presentation of the results would benefit*

from greater structure so the value of synthesis across the multiple SRs is clearer to the reader.

We thank the reviewer for the insightful reference; we were not previously aware of Prof Greenhalgh's work and read the reference with interest. We adopted the approach of a systematic review of reviews and used the PRISMA checklist. We aligned the reporting of results with the reviews included in our study, which broadly reported outcomes in the following sequence: clinical outcomes, patient reported outcomes, and use of health service resources. Hence, we believed that maintaining this overall structure would allow for synthesis across the various reviews in a manner that is better aligned with the existing HaH literature, and also facilitate understanding for readers who are familiar with current literature.

(2) *Discussion, p18. Line 56-58: Mortality of AA is compared with readmission of ESD. The reviewer thinks comparing the result of both mortality of AA and ESD is appropriate in this sentence.*

We noted this comment and have restructured the discussion accordingly (4. Discussion, paragraph 2).

(3) *Discussion, p19, line 5-8: The reviewer cannot understand the meaning of the sentence from the context, "However, apart from a study in one review suggesting this, insufficient evidence regarding adverse events precluded clear conclusion". Especially what "this" stand for?*

"This" was referring to the reduction of iatrogenic events under AA as compared to ESD. We have clarified this in the discussion as well. (4. Discussion, paragraph 2).

(4) *You name reduction of iatrogenic events such as delirium and infections as a benefit of AA-type over ESD-type HAH. In addition to this, there is less need for repeated patient data transfer between hospital units when the patient is directly admitted to a HAH instead of after spending some time on an inpatient ward. As different ICT-systems rarely work well together and each discharge possesses a risk of relevant information being lost, the fewer possibilities of human error the better.*

We thank the reviewer for this comment and have included it as part of the discussion. (4. Discussion, paragraph 6).

(5) *As mention above, the reviewer had an impression that LOS is highly important outcomes to assess HaH program's effectiveness. From the result and discussion, it is showed that total LOS in ESD and LOS (hospital or total?) in AA are increased. Is it a limitation of HaH? The authors are required to consider and discuss about this result more specifically. One of the opinions of the reviewer is that it would be interesting if the length and rate for complete cure of patients in hospital and HAH is compared.*

This is not a limitation of HaH. Some implementations (Tibaldi 2009; Ricauda 2004) had a longer LOS because the programme(s) included the subacute/rehabilitation phase into the same HaH episode— without which the patient may have gone on to other facilities (e.g.

community hospitals, skilled nursing facilities) to continue the care. These are not included in the LOS calculation for traditional hospitalisations. Some implementations (Mendoza 2009) also realised reduction in hospitalisation costs despite longer LOS due to tailoring of service intensity and reduction in fixed costs from traditional hospitalisations. We thank the reviewer for this question and have included these points into the discussion. (4. Discussion, paragraph 4)

Regarding the reviewer's suggestion on comparing LOS and rate of complete cure, the objective of many HaH episodes are not curative in nature. The principles of medicine in today's aging population and multimorbidity context have also shifted to one that is "To cure sometimes, to relieve often, to comfort always". As such, we will disagree that HaH is there to primarily "cure", but to provide care, and most importantly, comfort.

(6) *Discussion, about ESD/AA: Despite the authors showed the result of the reviews included both ESD and AA (third use case), there is no discussion about this use case. The authors also need to discuss about how you think and make implication about both included use case in Discussion.*

We have noted this comment and elaborated accordingly in the discussion. We would also like to take this opportunity to clarify that ESD/AA reviews are reviews that included both ESD and AA studies (i.e. the reviews still distinguished between these two use cases); they are not reviews that assessed interventions that are a hybrid of ESD and AA. (4. Discussion, paragraph 5)

(7) *The division of HAH models to ESD, AA and ESD/AA is understandable for scientific purposes, but in some case may be ill-suited for real-life settings. If an AA-type HAH is running at full capacity and is unable to admit a patient today, but can admit the same patient tomorrow (after the patient has spent one night at the hospital ward) is it actually a ESD/AA-type HAH then? A modern ESD/AA -type HAH can for example admit nursing home patients without moving the patient anywhere during the process. In general, the HAH field is in many aspects still very heterogenic and more subgroup analysis is needed in future studies.*

We acknowledge that this is a gap between research and actual implementation. In real-life settings, operational efficiencies and capacity will clearly come into the picture. However, we believe that the comparison between ESD and AA is important for scientific literature and also to provide a more focused direction for clinicians/ healthcare administrators who are new to the HaH model and keen to explore piloting HaH programmes. Nevertheless, we thank the reviewer for highlighting this pertinent point and have included this in our discussion as well. (4. Discussion, paragraph 7)

(8) *Discussion, p19-20 4.2 Recommendation for Practice: In this section, the authors are making discussion based on the previous literature rather than the result from this research. Since the discussion of HaH in the previous literature so far have been made from healthcare systems point of view, the reviewer thinks that writing the discussion based on the result of process indicator, facilitator and barrier will contribute as new opinion for HaH exploration.*

It was indeed one of our objectives to provide clarity on the process indicators, facilitators and barriers to HaH implementation that would be of interest to potential adopter organisations. Unfortunately, there was limited reporting of these indicators and factors in the included reviews. Hence, we focused the first section of our discussion on the primary outcomes reported in our results, whereas the second section on recommendations for practice was supplemented with sources beyond those in our study.

(9) *In facilitator and barriers section you write: "In addition, caregiver availability was an eligibility criteria in one-third to half of the studies across reviews". Does this mean availability of a caregiver if one needs him/her or availability absolutely? If the latter is true, please adress the question in discussion, is the HAH designed only for people who live with someone or in a nursing home etc?*

Yes. For some implementations, caregivers (formal/informal) were needed as part of the programmes to deliver effective care for the patients at home. For other implementations, such availability was either optional, or patients with high care needs were excluded from enrolment. We recognise that caregivers may not be universally available. As such, aspiring clinicians and hospital administrators should keep this factor in mind when designing their respective programmes along with patient inclusion/exclusion criteria. Nonetheless, we would also like to encourage the exploration of HaH programme designs which adopt appropriate safeguards to increase patient independence and reduce caregiver reliance. Further elaboration on this has been added to the discussion segment. (4.2. Recommendations for practice, paragraph 3)

Conclusion

(1) *Conclusion, p21, line 26-31: It seems that this first sentence in Conclusion is came from Introduction. The authors need to write here based on the result.*

(2) *Conclusion, p21, line 31-38: The reviewer think second and third sentence recommend to write transpose, to be like the logical order written in Conclusion in Abstract.*

We have noted both comments and amended the conclusion accordingly.

VERSION 2 – REVIEW

REVIEWER	Kimiko Mizuma Kagoshima University Graduate School of Medical and Dental Sciences, Japan
REVIEW RETURNED	12-Dec-2020

GENERAL COMMENTS	The revisions the authors made in the revised manuscript significantly improved the clarity, robustness, and strength of the author's argument and clarity of writing. I think this manuscript is relevant, topical, and deserves publication.

REVIEWER	Antti Ohvanainen Siun sote, Finland

REVIEW RETURNED	15-Dec-2020

GENERAL COMMENTS	Thank you for submitting your revised manuscript. I find it now more balanced and suitable for publishing.

REVIEWER	Natasha A. Lannin Monash University, Australia Declare that I am a co-author on one of the included systematic review which is synthesised in this meta-review.
REVIEW RETURNED	06-Dec-2020

GENERAL COMMENTS	OVERALL: I note the amended manuscript which has addressed many of the reviewer's earlier concerns. The topic remains of interest as the importance of considering alternate methods to provide acute medical care remains key (in particular within our COVID realities). Language and clarity would benefit from further editing with view to grammar and scientific expression, in particular there is repetition between sections and long sentences. This review was not registered prior to commencing, and I have noted some differences between manuscripts which I am unsure is simply addressing feedback or changing the method (See results table). The lack of registration prior to commencing and the potential biases should be made explicit in the abstract and main manuscript. . ABSTRACT: Objective continues to read oddly. Perhaps better reworded as: "To provide an overview of the safety and effectiveness of Hospital-at-Home (HaH) according to programme type (early-supported discharge (ESD) versus admission avoidance (AA)), and identify the model with highest evidence for addressing length of stay, readmission and cost outcomes." Noting, however, that the authors do not list these outcomes (rising costs or capacity constraints) hence I have assumed by rising costs, this matches to your lower cost outcome, and capacity constraints matches to your readmission/LOS outcomes. Results: Please cite your PRISMA data here, not just your total number of included. Please provide the range of OR for primary outcomes in summary, rather than textual synthesis only. Conclusion: Please use accepted scientific language with regards to findings of systematic reviews: I would suggest "largely results in" and "can potentially" may benefit from greater precision in expression. MANUSCRIPT: Results: Consistent with above suggestion for your abstract, please provide the OR in text for the synthesised statements: Eg. Line 14/15 page 16, you could easily add in the actual data from the Cochrane review. The key data for readers are not p-values but rather the ORs with their corresponding 95% CI. Please ensure that when you provide estimates, the measure of variability is also provided (eg. Line 14, p19) Discussion: While I appreciate the level of re-writing that has been done, the discussion is still unnecessarily long, and given the lack of data being presented in the results, largely repetitious of the results section. I would encourage the authors to reflect on the potential clinical implications of their review, and present these as succinctly as possible. For instance, the most important finding of this review is not that you believe it to be the first synthesis of its kind- surely your

	discussion point on line 35 of your discussion (that HAH is generally associate with similar or improved clinical outcomes, shorter LOS and higher levels of patient satisfaction) when compared to inpatient care is?

VERSION 2 – AUTHOR RESPONSE

Reviewer #1: The revisions the authors made in the revised manuscript significantly improved the clarity, robustness, and strength of the author's argument and clarity of writing. I think this manuscript is relevant, topical, and deserves publication.

Reviewer #2: Thank you for submitting your revised manuscript. I find it now more balanced and suitable for publishing.

We thank the reviewers for their kind acknowledgement and time spent on reviewing the manuscript.

Reviewer #3: I note the amended manuscript which has addressed many of the reviewer's earlier concerns. The topic remains of interest as the importance of considering alternate methods to provide acute medical care remains key (in particular within our COVID realities). Language and clarity would benefit from further editing with view to grammar and scientific expression, in particular there is repetition between sections and long sentences. This review was not registered prior to commencing, and I have noted some differences between manuscripts which I am unsure is simply addressing feedback or changing the method (See results table). The lack of registration prior to commencing and the potential biases should be made explicit in the abstract and main manuscript.

We thank the reviewer for the valuable comments. We have revised the manuscript to be more succinct and highlighted the limitation of the lack of protocol registration. We would also like to clarify that previous amendments were made in agreement with reviewers' feedback, rather than modifications to the method. We have since gone through the reviewer's additional comments and revised the manuscript accordingly.

Abstract

(1) *Objective continues to read oddly. Perhaps better reworded as: "To provide an overview of the safety and effectiveness of Hospital-at-Home (HaH) according to programme type (early-supported discharge (ESD) versus admission avoidance (AA)), and identify the model with highest evidence for addressing length of stay, readmission and cost outcomes." Noting, however, that the authors do not list these outcomes (rising costs or capacity constraints) hence I have assumed by rising costs, this matches to your lower cost outcome, and capacity constraints matches to your readmission/LOS outcomes.*

We thank the reviewer for the suggestion and have reworded the objectives accordingly. The reviewer rightly pointed out that we aimed to compare LOS and costs between the HaH models to arrive at recommendations with respect to capacity constraints and rising costs.

The only clarification we would like to make is that both clinical outcomes (mortality and readmissions) were of interest given that these would form the basis of any discussion of adoption by health systems. We acknowledge that this was unclear in our previous draft.

(2) *Results: Please cite your PRISMA data here, not just your total number of included. Please provide the range of OR for primary outcomes in summary, rather than textual synthesis only.*
We have noted this comment and included a summary of the effect measures where applicable. Unfortunately, it was challenging to provide our PRISMA data in view of the word limit.

(3) *Conclusion: Please use accepted scientific language with regards to findings of systematic reviews: I would suggest "largely results in" and "can potentially" may benefit from greater precision in expression.*
We have noted this comment and modified the language accordingly.

Results

(1) *Consistent with above suggestion for your abstract, please provide the OR in text for the synthesised statements: Eg. Line 14/15 page 16, you could easily add in the actual data from the Cochrane review. The key data for readers are not p-values but rather the ORs with their corresponding 95% CI. Please ensure that when you provide estimates, the measure of variability is also provided (eg. Line 14, p19)*
We have noted this comment and summarised the effect estimates reported across reviews as a range. However, we would like to note that this was only applicable when the same type of effect measure was used across reviews (e.g. either OR or RR). Confidence intervals were included if the effect measure was only reported by one review. With regards to Line 14, p19, we were unable to provide the confidence interval as this was not reported by the original author.

Discussion

(2) *Discussion: While I appreciate the level of re-writing that has been done, the discussion is still unnecessarily long, and given the lack of data being presented in the results, largely repetitious of the results section. I would encourage the authors to reflect on the potential clinical implications of their review, and present these as succinctly as possible. For instance, the most important finding of this review is not that you believe it to be the first synthesis of its kind- surely your discussion point on line 35 of your discussion (that HAH is generally associate with similar or improved clinical outcomes, shorter LOS and higher levels of patient satisfaction) when compared to inpatient care is?*
We have noted this comment and amended the discussion to be more concise. As the clinical implications of HaH have been extensively discussed in existing literature, we would like to clarify that we had greater interest in the policy implications and have taken the opportunity to highlight the types of health systems/ countries that would benefit from this model of care.